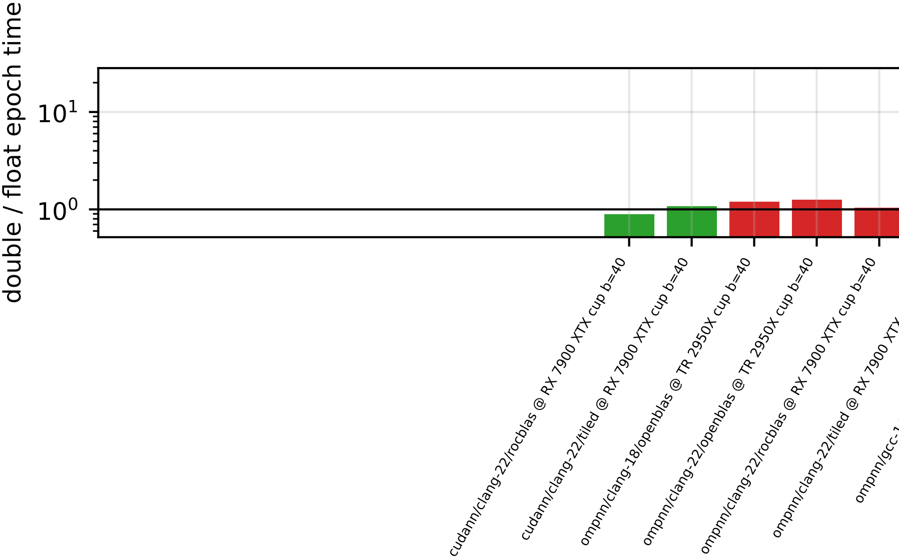
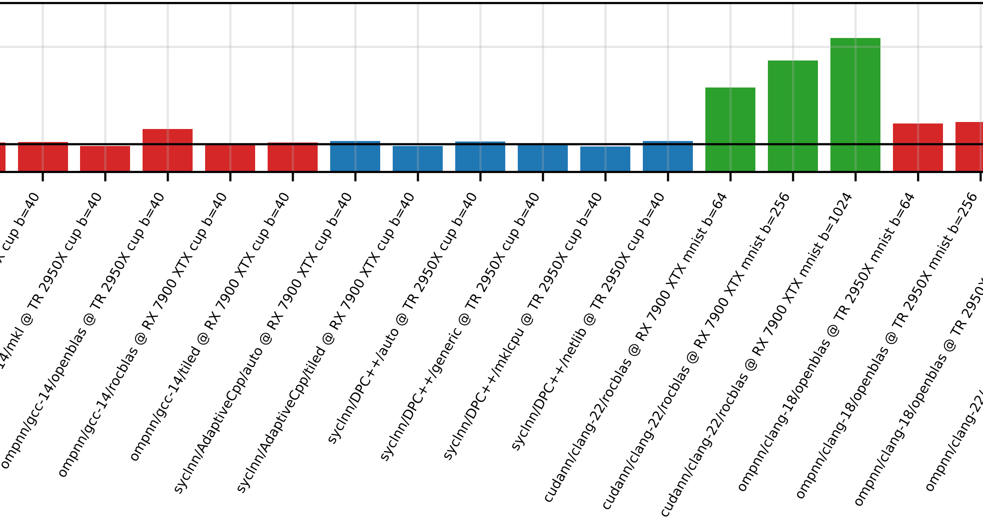
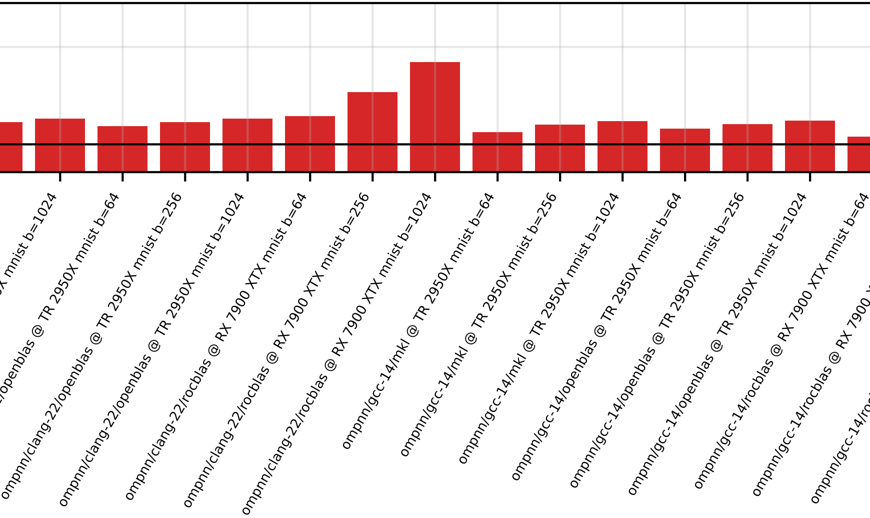
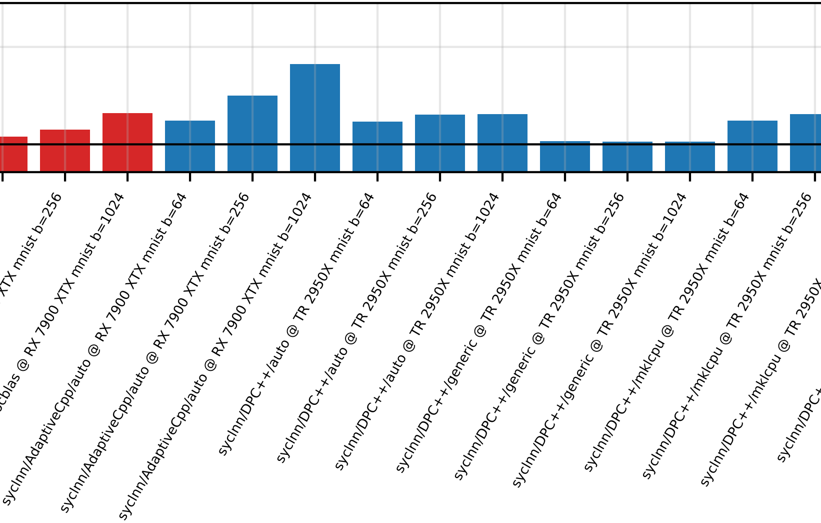
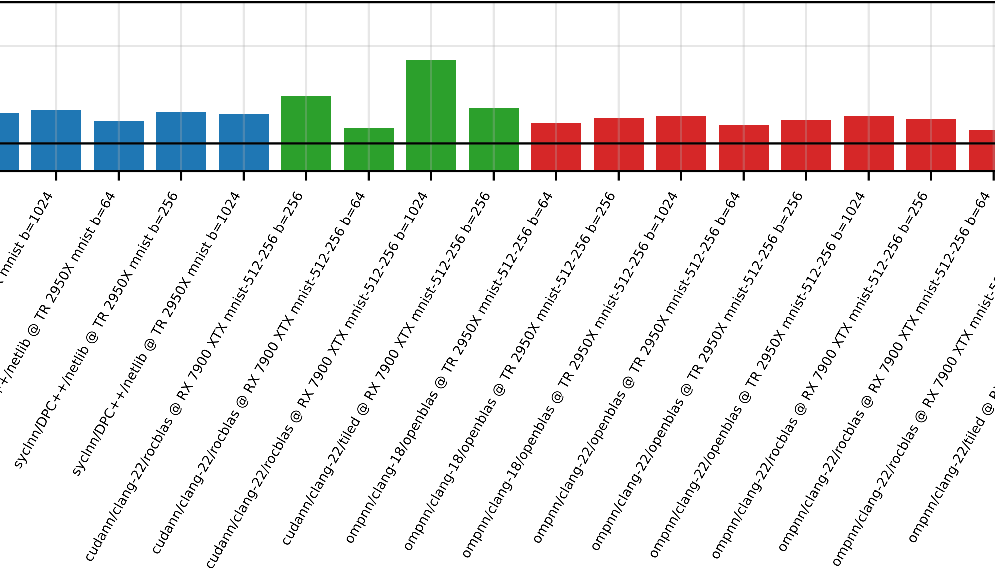
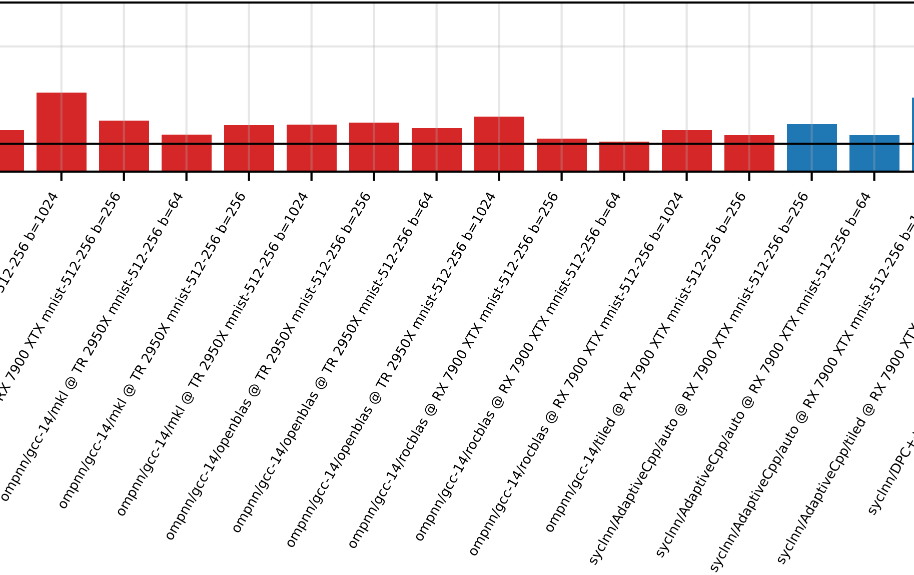

double / float epoch time
10^1
10^0
cudann/clang-22/rocblas @ RX 7900 XTX cup b=40
cudann/clang-22/tiled @ RX 7900 XTX cup b=40
ompnn/clang-18/openblas @ TR 2950X cup b=40
ompnn/clang-22/openblas @ TR 2950X cup b=40
ompnn/clang-22/rocblas @ RX 7900 XTX cup b=40
ompnn/clang-22/tiled @ RX 7900 XT
ompnn/gcc-

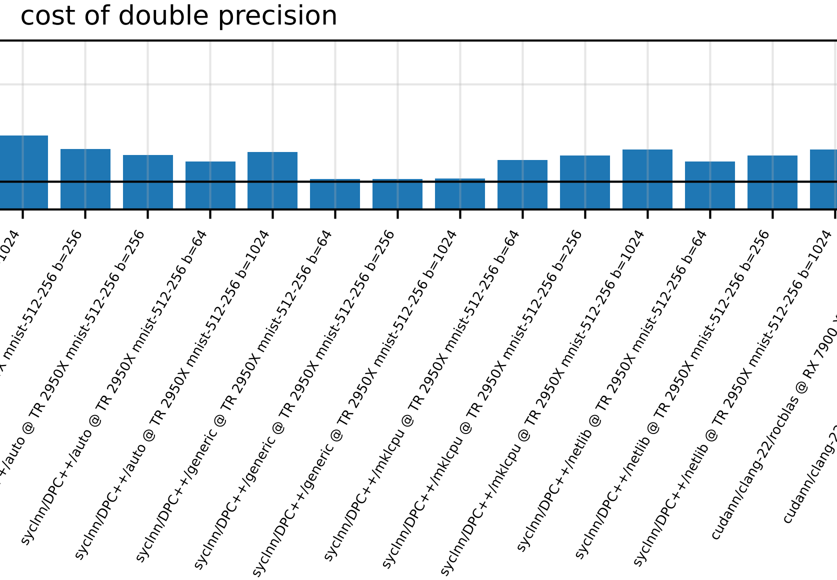

X cup b=40
14/mkl @ TR 2950X cup b=40
ompnn/gcc-14/openblas @ TR 2950X cup b=40
ompnn/gcc-14/rocblas @ RX 7900 XTX cup b=40
ompnn/gcc-14/tiled @ RX 7900 XTX cup b=40
syclnn/AdaptiveCpp/auto @ RX 7900 XTX cup b=40
syclnn/AdaptiveCpp/tiled @ RX 7900 XTX cup b=40
syclnn/DPC++/auto @ TR 2950X cup b=40
syclnn/DPC++/generic @ TR 2950X cup b=40
syclnn/DPC++/mklcpu @ TR 2950X cup b=40
syclnn/DPC++/netlib @ TR 2950X cup b=40
cudann/clang-22/rocblas @ RX 7900 XTX mnist b=64
cudann/clang-22/rocblas @ RX 7900 XTX mnist b=256
cudann/clang-22/rocblas @ RX 7900 XTX mnist b=1024
ompnn/clang-18/openblas @ TR 2950X mnist b=64
ompnn/clang-18/openblas @ TR 2950X mnist b=256
ompnn/clang-18/openblas @ TR 2950X
ompnn/clang-22/

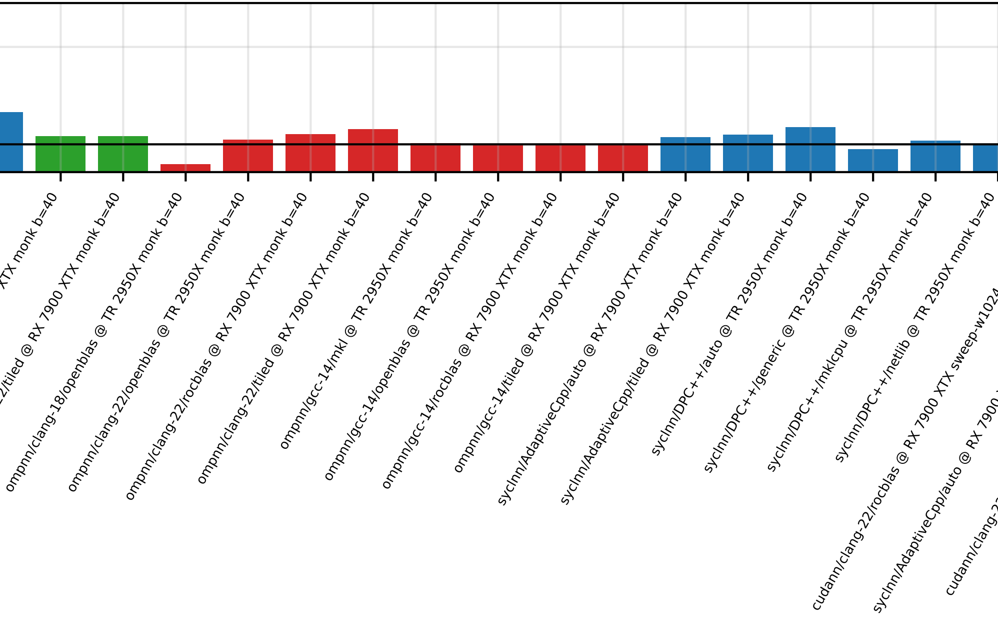

X mnist b=1024
/openblas @ TR 2950X mnist b=64
ompnn/clang-22/openblas @ TR 2950X mnist b=256
ompnn/clang-22/openblas @ TR 2950X mnist b=1024
ompnn/clang-22/rocblas @ RX 7900 XTX mnist b=64
ompnn/clang-22/rocblas @ RX 7900 XTX mnist b=256
ompnn/clang-22/rocblas @ RX 7900 XTX mnist b=1024
ompnn/gcc-14/mkl @ TR 2950X mnist b=64
ompnn/gcc-14/mkl @ TR 2950X mnist b=256
ompnn/gcc-14/mkl @ TR 2950X mnist b=1024
ompnn/gcc-14/openblas @ TR 2950X mnist b=64
ompnn/gcc-14/openblas @ TR 2950X mnist b=256
ompnn/gcc-14/openblas @ TR 2950X mnist b=1024
ompnn/gcc-14/rocblas @ RX 7900 XTX mnist b=64
ompnn/gcc-14/rocblas @ RX 7900 X
ompnn/gcc-14/ro

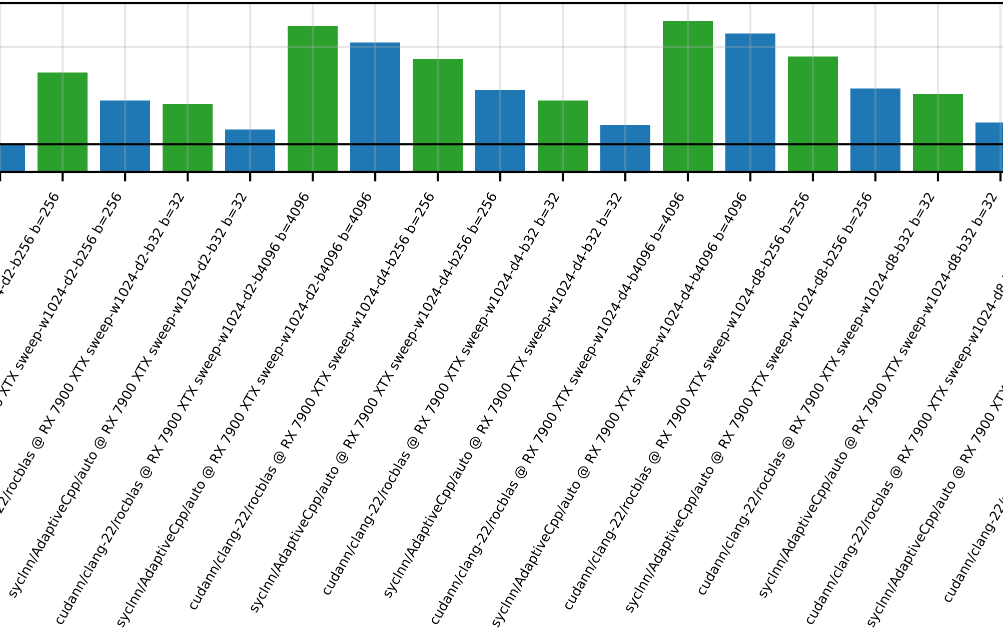

XTX mnist b=256
cblas @ RX 7900 XTX mnist b=1024
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist b=64
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist b=1024
syclnn/DPC++/auto @ TR 2950X mnist b=64
syclnn/DPC++/auto @ TR 2950X mnist b=256
syclnn/DPC++/auto @ TR 2950X mnist b=1024
syclnn/DPC++/generic @ TR 2950X mnist b=64
syclnn/DPC++/generic @ TR 2950X mnist b=256
syclnn/DPC++/generic @ TR 2950X mnist b=1024
syclnn/DPC++/mklcpu @ TR 2950X mnist b=64
syclnn/DPC++/mklcpu @ TR 2950X mnist b=256
syclnn/DPC+
TR 2950X

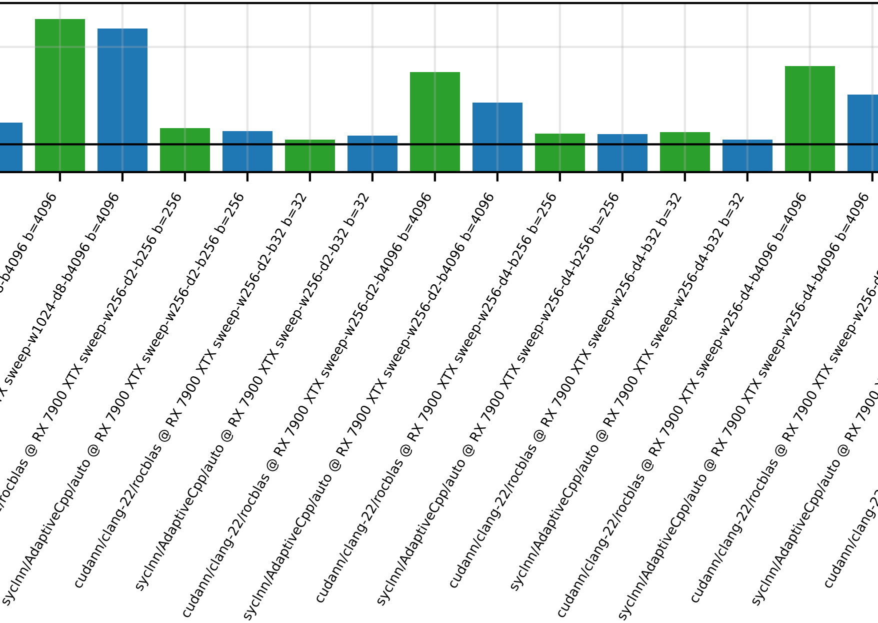

X mnist b=1024
+/netlib @ TR 2950X mnist b=64
syclnn/DPC++/netlib @ TR 2950X mnist b=256
syclnn/DPC++/netlib @ TR 2950X mnist b=1024
cudann/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=64
cudann/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=1024
cudann/clang-22/tiled @ RX 7900 XTX mnist-512-256 b=256
ompnn/clang-18/openblas @ TR 2950X mnist-512-256 b=64
ompnn/clang-18/openblas @ TR 2950X mnist-512-256 b=256
ompnn/clang-18/openblas @ TR 2950X mnist-512-256 b=1024
ompnn/clang-22/openblas @ TR 2950X mnist-512-256 b=64
ompnn/clang-22/openblas @ TR 2950X mnist-512-256 b=256
ompnn/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=1024
ompnn/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=256
ompnn/clang-22/rocblas @ RX 7900 XTX mnist-512-256 b=64
ompnn/clang-22/rocblas @ RX 7900 XTX mnist-5
ompnn/clang-22/tiled @ R

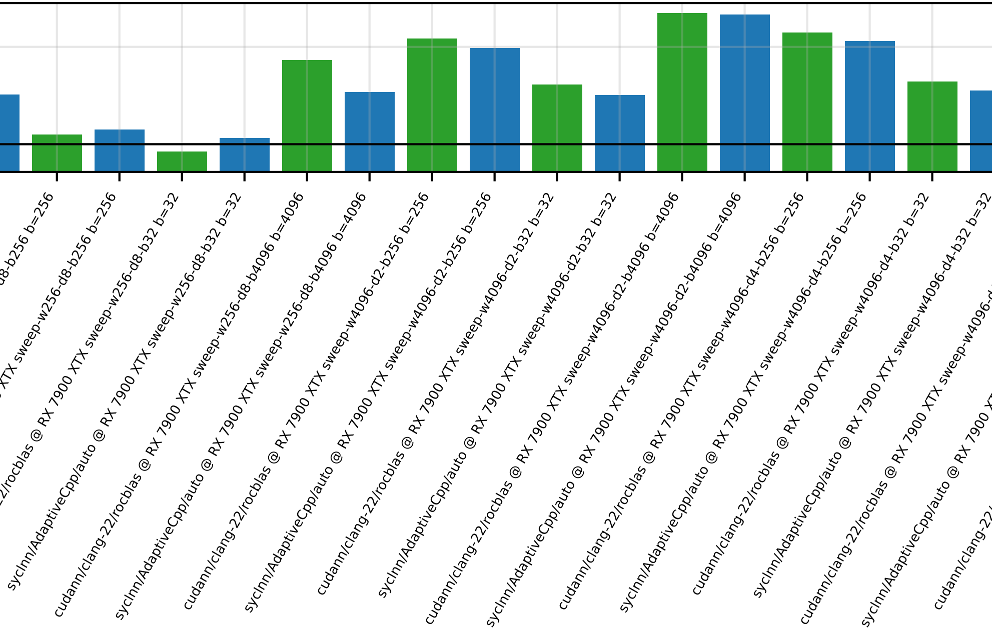

12-256 b=1024
RX 7900 XTX mnist-512-256 b=256
ompnn/gcc-14/mkl @ TR 2950X mnist-512-256 b=64
ompnn/gcc-14/mkl @ TR 2950X mnist-512-256 b=256
ompnn/gcc-14/mkl @ TR 2950X mnist-512-256 b=1024
ompnn/gcc-14/openblas @ TR 2950X mnist-512-256 b=256
ompnn/gcc-14/openblas @ TR 2950X mnist-512-256 b=64
ompnn/gcc-14/openblas @ TR 2950X mnist-512-256 b=1024
ompnn/gcc-14/rocblas @ RX 7900 XTX mnist-512-256 b=256
ompnn/gcc-14/rocblas @ RX 7900 XTX mnist-512-256 b=64
ompnn/gcc-14/rocblas @ RX 7900 XTX mnist-512-256 b=1024
ompnn/gcc-14/tiled @ RX 7900 XTX mnist-512-256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist-512-256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist-512-256 b=64
syclnn/AdaptiveCpp/auto @ RX 7900 XTX mnist-512-256 b=1
syclnn/AdaptiveCpp/tiled @ RX 7900 XT
syclnn/DPC+

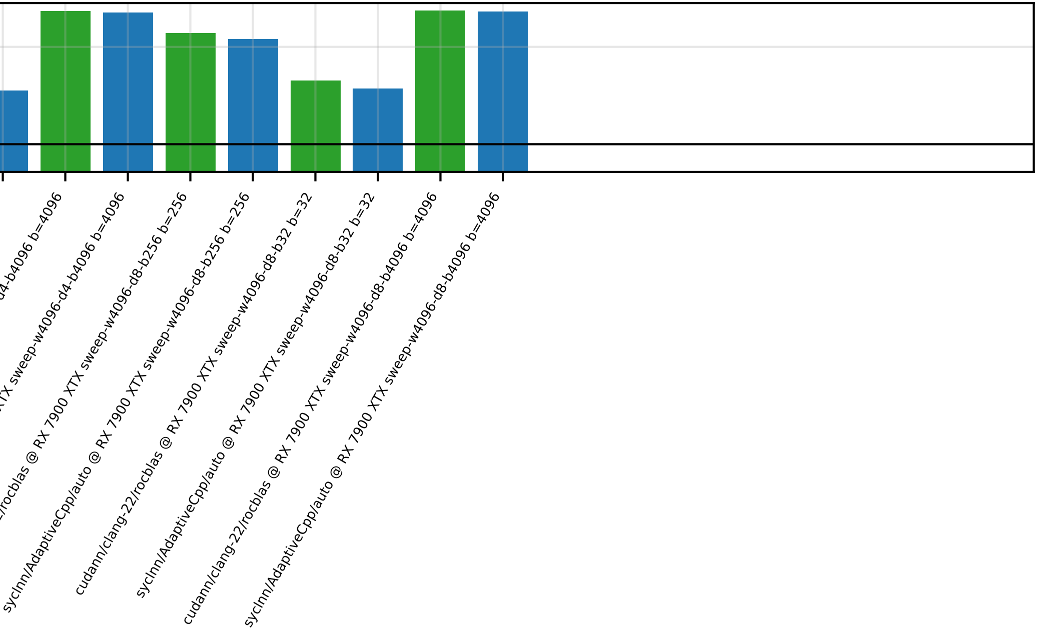

cost of double precision
1024
X mnist-512-256 b=256
+/auto @ TR 2950X mnist-512-256 b=256
syclnn/DPC++/auto @ TR 2950X mnist-512-256 b=64
syclnn/DPC++/auto @ TR 2950X mnist-512-256 b=1024
syclnn/DPC++/generic @ TR 2950X mnist-512-256 b=64
syclnn/DPC++/generic @ TR 2950X mnist-512-256 b=256
syclnn/DPC++/generic @ TR 2950X mnist-512-256 b=1024
syclnn/DPC++/mklcpu @ TR 2950X mnist-512-256 b=64
syclnn/DPC++/mklcpu @ TR 2950X mnist-512-256 b=256
syclnn/DPC++/mklcpu @ TR 2950X mnist-512-256 b=1024
syclnn/DPC++/netlib @ TR 2950X mnist-512-256 b=64
syclnn/DPC++/netlib @ TR 2950X mnist-512-256 b=256
syclnn/DPC++/netlib @ TR 2950X mnist-512-256 b=1024
cudann/clang-22/rocblas @ RX 7900 X
cudann/clang-2


XTX monk b=40
2/tiled @ RX 7900 XTX monk b=40
ompnn/clang-18/openblas @ TR 2950X monk b=40
ompnn/clang-22/openblas @ TR 2950X monk b=40
ompnn/clang-22/rocblas @ RX 7900 XTX monk b=40
ompnn/clang-22/tiled @ RX 7900 XTX monk b=40
ompnn/gcc-14/mkl @ TR 2950X monk b=40
ompnn/gcc-14/openblas @ TR 2950X monk b=40
ompnn/gcc-14/rocblas @ RX 7900 XTX monk b=40
ompnn/gcc-14/tiled @ RX 7900 XTX monk b=40
syclnn/AdaptiveCpp/auto @ RX 7900 XTX monk b=40
syclnn/AdaptiveCpp/tiled @ RX 7900 XTX monk b=40
syclnn/DPC++/auto @ TR 2950X monk b=40
syclnn/DPC++/generic @ TR 2950X monk b=40
syclnn/DPC++/mklcpu @ TR 2950X monk b=40
syclnn/DPC++/netlib @ TR 2950X monk b=40
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024
syclnn/AdaptiveCpp/auto @ RX 7900
cudann/clang-2


o XTX sweep-w1024-d2-b256 b=256
22/rocblas @ RX 7900 XTX sweep-w1024-d2-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d2-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d2-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d2-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d2-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d4-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d4-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d4-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d4-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d4-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d4-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d8-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d8-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d8-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w1024-d8-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w1024-d8
cudann/clang-22/


-b4096 b=4096
X sweep-
/rocblas @ RX 7900 XTX sweep-w1024-d8-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d2-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d2-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d2-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d2-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d2-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d2-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d4-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d4-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d4-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d4-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d4-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d4-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d
cudann/clang-2


d8-b256 b=256
XTX sweep-w256-d8-b256 b=256
22/rocblas @ RX 7900 XTX sweep-w256-d8-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d8-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w256-d8-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w256-d8-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d2-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d2-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d2-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d2-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d2-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d2-b4096 b=4096
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d4-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d4-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d4-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d4-b32 b=32
cudann/clang-22/


d4-b4096 b=4096
TX sweep-w4096-d4-b4096 b=4096
/rocblas @ RX 7900 XTX sweep-w4096-d8-b256 b=256
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d8-b256 b=256
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d8-b32 b=32
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d8-b32 b=32
cudann/clang-22/rocblas @ RX 7900 XTX sweep-w4096-d8-b4096 b=4096
syclnn/AdaptiveCpp/auto @ RX 7900 XTX sweep-w4096-d8-b4096 b=4096